

Table 1: Comparative Performance Evaluation of Conventional vs. Upgraded EPS Mortar

Comprehensive 2x2 Experimental Matrix (Mean Values with Standard Deviations)

Property / Performance Metric	Conventional EPS (20% Replacement)	Upgraded EPS (20% Replacement)	Conventional EPS (40% Replacement)	Upgraded EPS (40% Replacement)
Density, 28-day Oven-Dry (kg/m³)	2011.0 ± 18.5	1969.0 ± 17.0	1748.0 ± 15.0	1734.0 ± 14.5
Compressive Strength, fc (MPa)	10.15 ± 0.65	15.62 ± 0.85 (+53.9%)	4.20 ± 0.35	5.21 - 7.15 (+70.2%)
Flexural Strength, fr (MPa)	2.10 ± 0.15	3.59 ± 0.20 (+71.0%)	0.95 ± 0.08	1.80 - 1.85 (+94.7%)
Fracture Toughness (mJ/mm³)	12.58 ± 0.85	18.24 ± 1.20 (+45.0%)	1.87 ± 0.15	3.84 ± 0.25 (+105.3%)
Apparent Porosity (%)	7.65 ± 0.25%	6.48 ± 0.24%	8.03 ± 0.30%	6.94 ± 0.26%
Water Absorption (%)	3.82 ± 0.15%	3.34 ± 0.14%	4.50 ± 0.18%	3.52 ± 0.15%
ITZ Ca/Si Atomic Ratio (SEM-EDS)	23.50 ± 2.40 (CH)	6.80 ± 0.55 (C-S-H)	24.85 ± 2.70 (CH)	5.65 ± 0.39 (-77.3%)
Strength-to-Cost (MPa / 1000 LKR·m³)	0.275 ± 0.020	0.323 ± 0.025 (+17.5%)	0.098 ± 0.009	0.131 ± 0.011 (+33.7%)
Embodied Carbon (kg CO₂-e / MPa·m³)	53.47 ± 2.80	32.29 ± 1.60 (-39.6%)	130.02 ± 6.50	95.66 ± 4.50 (-26.4%)

*Note: Values represent mean ± standard deviation from 3 laboratory replicates. Upgraded mixes utilize PVAc coated EPS + 10% RHA + 0.5% PP fibers.